KANAK ANIL MALPANI

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PROFILE

Second-year B.Tech. student in Electrical and Electronics Engineering at NIT Tiruchirappalli, focused on efficient language-model adaptation, computer vision and agentic systems. Research and project interests span rigorous ML evaluation, on-device AI and accessible, practical engineering solutions.

EDUCATION

National Institute of Technology Tiruchirappalli Ongoing

B.Tech. — Electrical and Electronics Engineering

International Indian School, Dammam | Class XII, CBSE 2023–2025

Mayo College, Ajmer | Class X, CBSE 2021–2023

Relevant coursework (completed and in progress): Linear Algebra and Calculus; Fourier Transforms and Numerical Techniques; Signals and Systems; Circuit Theory; Complex Analysis and Differential Equations; Digital Electronics; Electron Devices.

PUBLICATIONS & RESEARCH

Efficient Adaptation of Small Language Models for Reliable Mathematical Reasoning

Kanak Anil Malpani. GroundLM Workshop, ACL 2026. Published. aclanthology.org/2026.groundlm-1.91/

Compared six PEFT methods; Qwen2.5-1.5B + PiSSA achieved 71.4% on GSM8K, exceeding the unadapted 7B baseline (68.5%).

Reported consistent method ranking across three base models (Kendall's $\tau = 1.0$) and up to 69.6% lower training VRAM using QLoRA.

What Does the Clear–Cloudy Gap Measure? A Controlled Audit of WorldFloods Flood Segmentation

CVPR 2027 submission — under review.

- Audited apparent cloud-related performance differences in WorldFloods v2 segmentation using scene-composition matching and same-tile synthetic cloud transplantation.
- Raw 39.3-point water-IoU gap across eight paired scenes decreased to 8.9 points after water-fraction matching and 1.4 points with stricter water-fraction/NDVI co-matching (five scenes); intervals were inconclusive for a matched cloud effect.
- Same-tile transplantation across 32 scenes reduced water IoU on still-visible pixels by 5.6 points; interpreted as synthetic boundary/adjacency sensitivity rather than proof of natural-cloud causality.

EXPERIENCE

Petronash | Electrical & Instrumentation Intern May 2026 · 1 month

- Studied industrial motors, DOL and star–delta starting, and Type-2 protection coordination.
- Reviewed MCCB/ELCB protection, thermal overload relays and LTMR motor-protection components.
- Read P&IDs and process-flow diagrams; studied electric heat-tracing systems and controls.
- Observed production and skid QC, pump factory-acceptance testing and interdepartmental project workflows.

COMPETITIONS & BUILD CHALLENGES

PEC Hacks | ₹60,000 prize | Code

ArogyaX (Team project): offline cardiac-screening prototype combining BLE ECG capture, on-device AI triage and encrypted records.

iQOO Hyderabad | 1st runner-up | ₹80,000 prize | Code

Nadaka: on-device walking-safety assistant for blind and low-vision users, with obstacle alerts, haptics and speech.

iQOO Bangalore | Grand finals — top 7 | Code

KHOJ: offline Bluetooth mesh for finding separated family members and routing life-safety alerts in dense crowds.

Forge (NxtLaunchpad) | Top 10 | Code

Saakshya: multilingual AI assistance that turns cybercrime accounts into checked complaint drafts and next-step guidance.

AWS BharatBuilds, Bangalore | Code

Absolute Latency Reduction: Amazon Bedrock streaming gateway and benchmark harness comparing caching, models and regions.

TECHNICAL SKILLS

Python, C | Machine learning, deep learning, computer vision, AI pipeline design | AI agents and multi-agent orchestration | Workflow automation and scripting | Drone assembly; embedded systems basics

SELECTED PROJECTS

Artificial-Emotions | Code

Exploring how to build an emotional layer within large language models.

Binary-Neural-Networks | Independent project | Code

Packed binary/ternary CPU inference using XNOR–popcount kernels; reports 32× weight compression and ~24× faster execution than NumPy FP32 on a 64×4096×4096 CPU benchmark.

Self-Learning-Vision | Independent project | Code

Local-first visual memory system for face detection, unknown-entity clustering and recognition; related work includes assistive smart-glasses identity recall.

PEFT | Independent project | Code

Small-language-model fine-tuning toolkit supporting six PEFT methods, GSM8K/MATH evaluation and systems profiling.

LoopBench | Independent project | Code

Benchmark suite and scoreboard for feedback-driven systems, using fixed tasks and seeds.

Autonomous-Startup-in-a-Box | Code

AMD Developer Build Challenge project: six parallel agents coordinating startup ideation, research, product planning, branding, strategy and execution.

AI for Lawyers

Developing an AI tool for lawyers; currently in MVP testing.

Drone Build — NIT Trichy Technical Fest

Designed, assembled and configured a functional drone within 20 days; reached the competition semi-finals.

LEADERSHIP & SERVICE

Coordinator — E-Cell, NIT Trichy

Building a unified knowledge base for E-Cell members to support continuous improvement and effective knowledge sharing.

Coordinator — Technical Council, NIT Trichy | Second year

Oversees technical and managerial activities; helps host TransfiNITTe, a multi-company competition at NIT Trichy.

Marketing Team — Institute Fest & TEDx Club

Automated institute-fest stakeholder outreach and secured four clients in the first week; contributes to TEDx outreach and event promotion.

Prakriti — Member, 2 years | NSS — Member, 1 year

Supports campus animal and plant protection through Prakriti, and social hygiene and cleanliness awareness through NSS.

AWARDS & COMPETITIONS

1st place — Inter-Institute Finance Event: Led a team through a simulated startup lifecycle, including crisis management, growth strategy and M&A negotiation.

3rd place — Finance & Strategy Event: Delivered competitive strategic analysis.

Finalist — Case Competition: Proposed an automobile-company pivot in response to changing consumer behaviour and market dynamics.

LANGUAGES

English (Fluent) | Hindi (Native) | Gujarati (Basic, speaking)
German (Learning, ongoing)